

AI Operations (AIOps) Q1

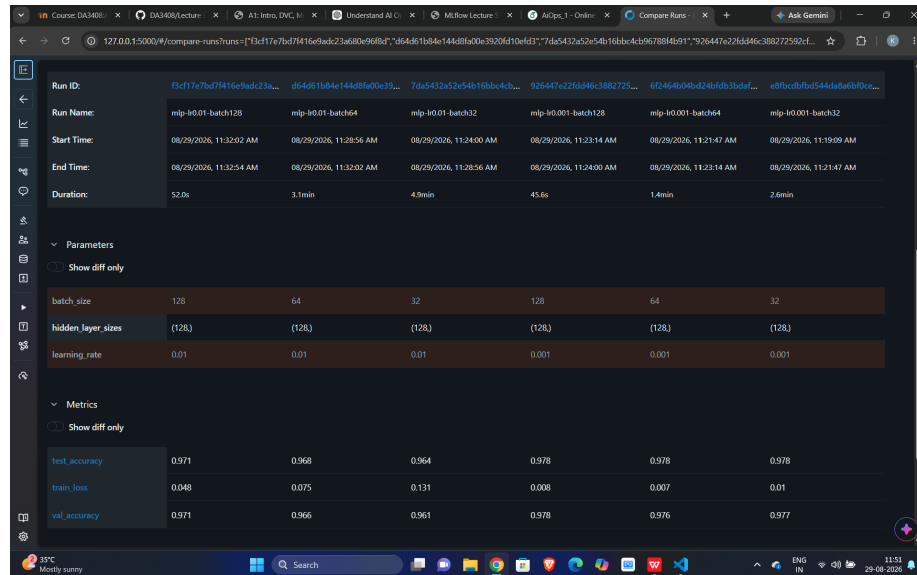
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Question 2 – MLflow Experiment Comparison

1. MLflow Run Comparison

The MLflow comparison of all six runs is shown below.



Run ID:	03d11767bd7f416e9ad23a...	d64d61b84e144d8f0a0e35...	7de5432a52e54b16bb04b...	926447e2256446c3832723...	6f2464b04d424bdfb3b0af...	e8fbc08f0a544da8a0b0c...
Run Name:	mip-0.01-batch128	mip-0.01-batch64	mip-0.01-batch32	mip-0.001-batch128	mip-0.001-batch64	mip-0.001-batch32
Start Time:	08/29/2026, 11:32:02 AM	08/29/2026, 11:28:56 AM	08/29/2026, 11:24:00 AM	08/29/2026, 11:23:14 AM	08/29/2026, 11:21:47 AM	08/29/2026, 11:19:09 AM
End Time:	08/29/2026, 11:32:54 AM	08/29/2026, 11:32:02 AM	08/29/2026, 11:28:56 AM	08/29/2026, 11:24:00 AM	08/29/2026, 11:23:14 AM	08/29/2026, 11:21:47 AM
Duration:	52.0s	3.1min	4.9min	45.6s	1.4min	2.6min
Parameters						
batch_size	128	64	32	128	64	32
hidden_layer_sizes	(128)	(128)	(128)	(128)	(128)	(128)
learning_rate	0.01	0.01	0.01	0.001	0.001	0.001
Metrics						
test_accuracy	0.971	0.968	0.964	0.978	0.978	0.978
train_loss	0.048	0.075	0.131	0.008	0.007	0.01
val_accuracy	0.971	0.966	0.961	0.978	0.976	0.977

Figure 1: MLflow comparison of the six MLP runs

2. Written Analysis

The best-performing run was the MLP with a learning rate of 0.001 and a batch size of 128. It achieved a test accuracy of 0.978 and a validation accuracy of 0.978, which was the highest validation accuracy among the six runs. Although the other two runs with a learning rate of 0.001 also achieved a test accuracy of 0.978, the batch size of 128 gave the best validation accuracy.

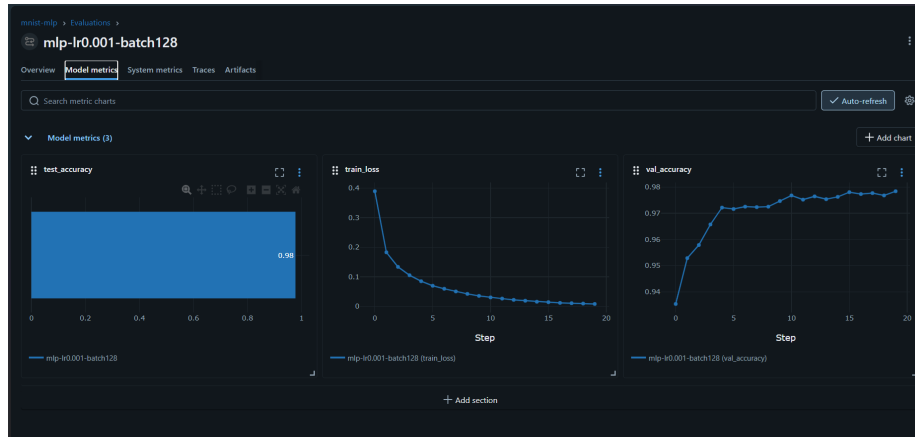


Figure 2: Training loss and validation accuracy for the best-performing run

There is some limited evidence of overfitting in the later epochs. The training loss continuously decreases from around 0.39 to 0.008, while the validation accuracy increases initially and then mostly stays around 0.975–0.978. Since the validation accuracy does not continuously decrease, there is no strong overfitting, but the continued decrease in training loss after the validation accuracy levels off suggests some overfitting.

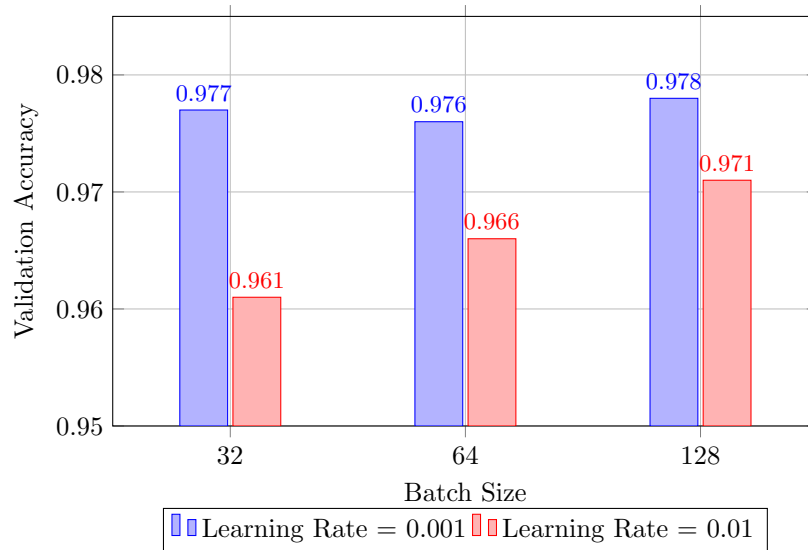


Figure 3: Validation accuracy across different learning rates and batch sizes

The learning rate appears to have a larger effect on performance than the batch size. Changing the learning rate from 0.001 to 0.01 reduced the validation accuracy for all three batch sizes. With a learning rate of 0.001, the validation accuracy remained between 0.976 and 0.978, whereas with 0.01 it was between 0.961 and 0.971. Therefore, the learning rate had the more noticeable effect on model performance.

3. MLflow Logging Code

The following code was added to log the experiment parameters and metrics:

```
mlflow.log_param("learning_rate", learning_rate)
mlflow.log_param("batch_size", batch_size)
mlflow.log_param("hidden_layer_sizes", "(128,)")

mlflow.log_metric("test_accuracy", test_accuracy)

for epoch, loss in enumerate(model.loss_curve_):
    mlflow.log_metric("train_loss", loss, step=epoch)

for epoch, val_acc in enumerate(model.validation_scores_):
    mlflow.log_metric("val_accuracy", val_acc, step=epoch)
```